

16.4 Green's Theorem

1 Warm Up

2 Motivation

In the last section we looked at what happens when we go to take a line integral over a vector field and my vector field is conservative. This took care of a special case of the line integral problem and with the FToC we solved it! What if instead of taking a special vector field and any curve, we reversed that and looked at any vector field and a specific curve? Can we also answer the line integral problem?

3 Definitions & Theorems

1. A closed simple curve is one that starts and ends at the same point and does not cross over itself. We also assume we travel in the counter-clockwise direction.
2. Green's Theorem says the following

Theorem 1. (*Green's Theorem*) If C is a simple closed curve containing the region R then,

$$\oint_C F \cdot dr = \oint_C P dx + Q dy = \iint_R \left[\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right] dA.$$

3. Just to make sure we see the various forms of a line integral and our notation is consistent, let $F(x, y) = P\hat{i} + Q\hat{j}$ and let our curve be defined by $\vec{r}(t) = x\hat{i} + y\hat{j}$. Remember $\vec{r}'(t) = f_x\hat{i} + f_y\hat{j}$. Then $F \cdot \vec{r}' = Pf_x + Qf_y$
4. Okay what if we had the vector field $F(x, y) = -y\hat{i} + x\hat{j}$. For now this just seems like a regular vector field. But now consider the line integral through through a closed loop over this vector field! (Well half of it)

$$\frac{1}{2} \oint_C -y dx + x dy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA = \frac{1}{2} \iint_R 1 - (-1) dA = \iint_R 1 dA.$$

We now have an equation for the area bounded by a closed simple curve in terms of line integrals! This is cool!

4 Examples

Problem 1. Evaluate the following line integral where C is the closed curve around the triangle with vertices $(0, 0)$, $(1, 1)$, $(0, 1)$.

$$\oint_C x^3 \, dx + xy \, dy$$

Problem 2. Evaluate the following line integral where C is the closed curve around $y = x$ and $y = x^2$.

$$\oint_C (x^2y + x^3) \, dx + 2xy \, dy$$

Problem 3. Evaluate the following line integral.

$$\oint_C x e^x \, dx + y^5 \, dy$$

Problem 4. Evaluate the following line integral where C is the closed curve around the top half semicircle $x^2 + y^2 = 1$ and the x -axis.

$$\oint_C x^2 y \, dx + y^3 \, dy$$

Problem 5. Evaluate the following line integral where C is the closed curve around the circle with radius 2.

$$\oint_C 6xy \, dx + (3x^2 + \ln(1+y)) \, dy$$

Problem 6. Find the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Problem 7. Find the work done through the vector field $F(x, y) = \frac{y}{1+x^2}\hat{i} + (x + \tan^{-1}(x))\hat{j}$ along the right loop of the curve $r^2 = \cos(2\theta)$.

5 Scratch Work